

FIXED DRUG ERUPTION – MOST COMMON ADVERSE DRUG REACTION

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ABSTRACT

This study was conducted from January 2001 to January 2002 to report adverse reactions of commonly used drugs to UPPSALA/SWEDEN, through CPSP approved by Ministry of Health. Adverse drug reaction is any unintended or undesired response to a medication given at an appropriate dose (Anderson, 1987). While reporting the adverse drug reactions, 50 cases out of 69 were diagnosed as having drug eruption (FDE). This is one of the most common cutaneous or mucocutaneous reaction that has been reported for numerous drugs. The incidence can occur at any time without positive previous history but can be revoked by repeating the use of the same drug and also chemically related substance (Matasin, 1993).

In our study maximum cases were seen as (FDE) with vibramycin (Doxycycline) next by the Septran (co-trimoxazole). These results coincide with various other studies conducted through-out the world. This also indicates neither genetic nor racial predisposition for developing FDE. The reaction can be minimized by having a proper knowledge of the offending substance and clinical experience of the lesions.

INTRODUCTION

Drugs are potent chemicals that often have effects besides the desired actions. These undesired actions may vary from being just negligible to severe anaphylaxis broadly based upon pharmacological, immunological, genetic and racial basis.

Adverse drug reactions account for 2% to 6% of the hospital admissions (Babu Geeta Belgi, 2002). Among the adverse drug reactions where skin becomes the target, FDE (Fixed Drug Eruption) is one of the most common presentation as well as most misdiagnosed condition by non-skin doctors (Ordaqu *et al.*, 1995). Cutaneous or mucutaneous eruptions are the most common adverse reaction to orally or parenterally introduced drugs and spectrum ranges from mild transitory exanthematous rash to toxic epidermal necrolysis (Chan *et al.*, 1984), potentially a fatal condition. Fixed drug exanthema is a disease secondary to drugs in 85%-100% of the cases and usually has

cutaneous eruption (Kauprinen, 1985).

Numerous drugs have been reported to produce FDE, but tetracyclines, sulfonamides, phenophthalines and some NSAIDS are the most common groups of the drugs producing it (Chan *et al.*, 1984). These are also the drugs which are commonly used by general practitioners. Less likely to produce reaction are digoxin ferrous sulphate and potassium chloride.

The typical cutaneous eruption consists of erythemoviolacious macule on the skin but mucosal involvement without skin lesion is also possible (Sharma *et al.*, 1996). A typical FDE may also occur in the form of bullae, ulcer or some other form Chan *et al.*, 1984, described three cases of lichenoid eruption linked with the administration of allopurinol. In these patients the withdrawal of drug resulted in clinical resolution of the oral mucosal lesions. The FDE lesions typically occur by offending drug or chemical in 1-3 number, commonly over hands, feet, glans

penis, lips and trunk, also can occur as early as 30 minutes and number increases with successive exposures (Mehboob and Haroon, 1998). Lesions fade in a week time and leave hyper pigmented patch that may last for several months. Similar type of reactions have been reported by chemically dissimilar drugs or agents (Matasin *et al.*, 1993). However there may be phases during which the person does not develop such reaction, indicating refractory period.

Immunological and non-immunological mechanisms are involved but how FDE exactly arises and in particular why just same area is involved has not been worked out clearly. FDE can occur at any time, any one must not assume that the patients have been taking the same drug for a long time.

Physicians and other practitioners should always consider the drug as possible causes of skin eruption in their patients. Throughout review all potential medications, prescriptions, alternations need to be scrutinized thoroughly if identification is to be achieved. The definitive diagnosis can be made by re exposure to the susceptible agent but this is contraindicated due to severity of reaction in rare cases (Ozkaya, 2000). Recently a linear

form of eruption has been reported as FDE by piroxicam sublingual pills (Ozkaya, 2000).

Generalized FDE eruption can happen in rare cases and has been reported by the use of phenazone chemicals and barbiturates (Ozkaya, 2000). Some studies also report co-trimoxazole and phenytoin as a cause of generalized FDE. Other reports claim co-trimoxazole and dipyrone as a cause of generalized FDE (Subb *et al.*, 1994).

Some studies report co-trimoxazole and dipyrone as uncommon cause for generalized FDE while other studies has reported co-trimoxazole and phenytoin as common cause of generalized FDE (Sharma and Thar, 1996).

OBJECTIVE

To establish a good safety profile of drugs used in Pakistan in association of adverse drug reaction (ADR) monitoring program approved by Ministry of Health under the department of CPSP (College of Physician & Surgeon) Pakistan Karachi as an associate member of WHO collaborating centre for international drug monitoring at UPPSALA/ Sweden.

Table-1

S.No.	Offending drug/agents	Total No. Patients	M	F	No. of Patients below 12 years of age
1.	Doxycycline	17	9	8	0
2.	Co-trimaxazole (Septran)	13	7	6	0
3.	Buscopan compositum	6	4	1	1
4.	Naproxin	3	2	1	0
5.	Gelatin	3	2	1	0
6.	Brufen	4	2	2	0
7.	Amoxil	1	0	0	1
8.	Calpol (Paracetamol)	1	1	0	0
9.	Unidentified	2	0	1	1
		50	27	20	3

PATIENTS & METHODS

The sufferers were followed for the recovery period and detailed previous history was obtained. All of them were also prescribed symptomatic remedies for the relief of their agony.

RESULTS

Out of a total of 64 cases 50 had FDE while remaining 14 had adverse drugs reactions other than FDE cases 49 developed eruptions involving the typical areas, palms feet trunk lips and glans penis. The lesions developed within 24 to 48 hours in the form of dark pigmented macules with burning sensations. Lesion were 1 to 3 in number. One case developed pigmented, ulcerated lesion in oral cavity involving the gingival mucosa and was not diagnosed till the offending drug (Welcodox) doxycycline was incidentally withdrawn and the patient got relief (Table-1).

As regards the adverse drug reactions other than FDE a total of 14 cases were evaluated (13 +1 infant). Majority of patients had extrapyramidal effects (6 patients) followed in number by 3 cases each of erythema multiform and purpura. 2 cases one each presented with vasoragal shock and anaphylaxis. Extrapyramidal effects were seen with promethazine, trifluoperazine and metaclopramide. The reaction involved the involuntary movements of the facial muscles and was controlled by intra muscular injection of diazepam.

One breast-feed developed action similar to when she given trifluoperazine (stelazine) one mg b.i.d. The effect was confirmed by repeating the same drug to the mother after one week. Chloroquine injection produced vasovagal shock and erythema multiform. Purpura was seen with aspirin. In addition to FDE co-trimoxazole produced erythema multiform. Lastly ATS injection caused anaphylaxis (Table-2).

Table-2

Offending Drug	V.V. Shock	Anaphyl-axis	Ext.Pyramidal Effects	E.M.	Purpura
Stelazine (Trifluoprazine)	-	-	2	-	-
Phenergan (Promethazine)	-	-	3	-	-
Maxalon (Metachlopramide)	-	-	1	-	-
Erythromycin	-	-	-	-	-
Methotrexate	-	-	-	-	-
Diclofenac Sodium	-	-	-	-	-
Cloroquine (Injection)	1	-	-	1	-
Aspirin	-	-	-	-	3
Septran	-	-	-	2	-
A.T.S. Injection	-	1	-	-	-
	1	1	6	3	3

V.V. = Vasovagal, E.M. = Erythema Multiform

DISCUSSION

In our study the highest number of FDE cases were seen with doxycycline (Vibramycin) followed next by the co-trimoxazole. Various studies indicate the highest incidence with co-trimoxazole (Ozaka 2000 and Stubb *et al.*, 1994). However both of these drugs are well established to produce FDE. In our study only one case below 12 years is mentioned, by amoxicillin (amoxil) probably due to lack of use of these two common drugs in young age group. Betalactam antibiotics and anticonvulsants are labeled to produce FDE in children under one year of age in a study conducted between 1992 to 2002). The highest incidence was found in 1st year of life by the betalactam antibiotic and anticonvulsant drugs in male sex (Barron, 1995).

Two unidentified cases were suffering from upper respiratory tract infections probably of viral origin, during their therapy with antibiotics and antipyretics. They developed exanthematous eruption which is also a manifestation of various viral diseases. As they were taking more than one susceptible drug out so definite conclusion could not be made.

Skin lesions were managed by local application of 1 % hydrocortisone cream along with oral use of antihistamines (Piriton of GSK) for 2 to 3 days. In our study 3 cases of FDE are also reported by Gelatin cover of Ampicilin Capsules. The reaction did not occur when the patient was given Ampicilin in powder form. In literature certain legumes, fruits, foods additives, gelatin and even U.V. radiation has been implicated (Barron, 1995).

None of the FDE cases needed hospitalization. Similar results in various studies through out the world indicated no genetic or racial predisposition in development of FDE. In the last 5 years study of the most common types of adverse reactions were fixed eruption, exanthematous eruption and urticaria, the most common groups were

antimicrobial, antipyretics, anti-inflammatory and analgesics (Barron, 1995). Our results are fully consistent with these findings.

During the last 17 years (1982 to 1986) 3561 cases of "drug eruption" have been reported in the major journals of dermatology in Japan, the number of patient per year showed a steep increase since 1992, the age distribution gradually shifted to the elderly in the last 10 years (Barron, 1995).

CONCLUSION

The incidence of FDE is increasing with the increasing number arid types of drugs and chemicals. Previous history of the patients and knowledge about the offending agent is necessary to make the diagnosis. By avoiding the common susceptible agents the incidence of FDE can be minimized.

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